PROJECT MODULE 4: VALIDATION SCRIPTING

Form Validation Scripting & Technical Report

Subject: Web Technology | Script Language: JavaScript (ES6+) | Date: 14-Oct-2026

1. AIM & OVERVIEW

The objective of client-side validation is to ensure strict data integrity before form submission. Numeric fields block alphabetical characters on keypress, name fields block numerical characters, and structured formats (Email, Pincode, Phone) are validated with Regular Expressions.

2. VALIDATION RULES MATRIX

Field Name	Rule Enforced	Allowed Input	Blocked Characters	RegEx Pattern
Patient Name	Alpha Only	Letters (a-z, A-Z), Space	Numbers (0-9), Symbols	/^[a-zA-Z\s.'-]+$/
Mobile Phone	Numeric + Special	Digits (0-9), +, -, Space	Alphabets (a-z, A-Z)	/^[\d\s+\-()]{10,15}$/
Pincode	6-Digit Exact	Only Digits (0-9)	Letters, Special Characters	/^\d{6}$/
Email ID	RFC 5322 Standard	user@domain.com	Missing @ or domain dot	/^[^\s@]+@[^\s@]+\.[^\s@]+$/
Patient Age	Integer Range	0 to 125	Negative, Alpha, >125	0 <= age <= 125

3. SCRIPTING LOGIC & CODE SNIPPETS

```
// A. Real-time Keystroke Blocker for Phone Numbers
function restrictPhone(event) {
if (['Backspace','Tab','ArrowLeft','ArrowRight','Delete'].includes(event.key)) return;
if (!/[\d+\-\s]/.test(event.key)) {
event.preventDefault(); // Block non-numeric keystroke
showShake(event.target, 'Alphabets blocked in phone number!');
}
}

// B. Real-time Name Field Digits Blocker
function restrictName(event) {
if (/[0-9]/.test(event.key)) {
event.preventDefault(); // Block number
showShake(event.target, 'Numbers blocked in name field!');
}
}
```

4. TESTING VERIFICATION & RESULT

Validation was rigorously tested in Chrome and Edge. Keystroke listeners successfully prevent invalid characters from appearing in form inputs. Field shaking animations and red highlights provide immediate visual feedback. Advance to subsequent steps is restricted until all step inputs pass validation.